

Plant Location



- Plant location refers to the **choice of region and the selection** of a particular **site** for setting up a business or factory.
- The choice is made only after considering **cost and benefits** of different alternative sites.
- It is a strategic decision that **cannot be changed once taken**. If at all changed only at considerable loss, the location should be selected as per its own requirements and circumstances.

Plant Location..



- The **purpose** of location study is to **find an optimum location** that will result the greatest advantages to the organization.
- Before the location of plant is selected, **long range forecast** should be made anticipating future need of the company.

Need for selecting suitable location

Need for selecting suitable location arises because of two situation:

1. In case of existing factory
2. When starting a new factory

The existing firms will seek new locations in order to expand the capacity or to place the existing facilities. The increase in demand for the company's products can give rise to following decisions:

- ✓ Whether to expand an existing capacity and facilities.
- ✓ Whether to look for new location for additional facilities.
- ✓ Whether to close down existing facilities to take advantage of some new location.

Need for selecting suitable location..



Reason for replacement of existing facilities to new location

- Change in location of demand
- Change availability of material
- Change availability of Transport
- Change in cost or supply of labor
- Change in regulation and law
- Change in policies of industries to relocate on which the firm is dependent

Importance of Location

Competition

- A company's location affects its ability to compete. In manufacturing firms, location affects direct cost by, to and from transportation from the location and cost of labour and others. Thus a good location helps to deliver the product at a cheaper price and hence helps to combat competition.

Cost

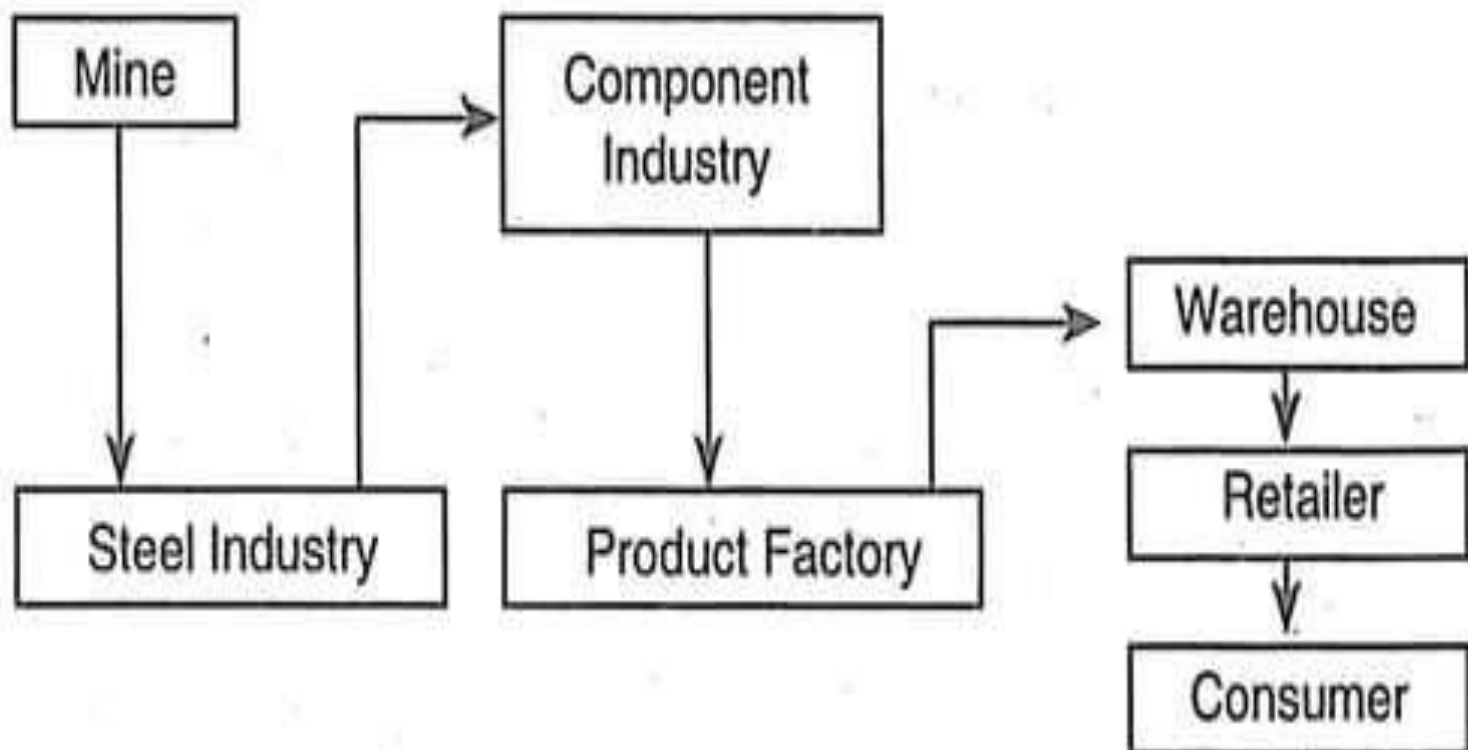
- Failures to make a good location become too expensive for the company and have long lasting effects. Plant location is a major investment decision. If selected wrongly, it is going to affect the performance of the company and location once selected will be difficult to change in near future.

Indirect benefits

- These are not visible directly and will not be reflected in company's accounts.

System view of Location

- The systems view evaluates all the components and their interrelationships to arrive at an optimal location for all the components of the chain.



Factor influencing to location/Considerations in Plant Location



1. Proximity to market

- Cost of distribution is an important item in the overhead expenses. So it will be advantageous to be near to the center of demand for finished products.

2. Supply of raw material

- One of the most important considerations involved in selection of industrial location has been the availability of raw materials required.
- The biggest advantage of availability of raw material at the location of industry is that it involves less cost in terms of transportation cost.

Factor influencing to location/Considerations in Plant Location



3. Transport facilities

- Since freight charges of raw materials and finished goods enter into the cost of production, therefore transportation facilities are becoming the governing factor in economic location of the plant.

4. Infrastructure availability

- The degree of dependency upon infrastructural facilities may vary from industry to industry, yet there is no denying of the fact that availability of infrastructural facilities plays a deciding role in the location selection of an industry.
- The infrastructural facilities include power, transport and communication, water, banking, etc.

Factor influencing to location/Considerations in Plant Location



5. Labour and wage

- Potential supply of requisite type of labour governs plant location to major extent.
- Some industries need highly skilled labour while other need cheap labour.

6. Law and taxation

- Laws prohibit the setting up of polluting industries in prone areas particularly which are environmentally sensitive.
- The taxation on a higher rate may discourage some industries from setting up in an area, the same in terms of tax holidays for some years may become the dominant decisional factor for establishing some other industries in other areas.

Factor influencing to location/Considerations in Plant Location



7. Suitability of lab and climate

- Topography, area, the shape of the site, cost, drainage and other facilities, the probability of floods, earthquakes (from the past history) etc., influence the selection of plant location.
- Climatic conditions vary from place to place in any country including India. And, climatic conditions affect both people and manufacturing activity. It affects human efficiency and behaviour to a great extent.

8. Supporting industries and services

- Now a day the manufacturing organization will not make all the components and parts by itself and it subcontracts the work to vendors. So, the source of supply of component parts will be the one of the factors that influences the location.

Factor influencing to location/Considerations in Plant Location



9. Community and labour attitude

- Community attitude towards their work and towards the prospective industries can make or mar the industry. Community attitudes towards supporting trade union activities are important criteria.
- Facility location in specific location is not desirable even though all factors are favoring because of labor attitude towards management, which brings very often the strikes and lockouts.

Factor influencing to location..



<i>Location Factor</i>	<i>Territory Selection</i>	<i>Community Selection</i>	<i>Site Selection</i>
1. Proximity to market	✓		
2. Supply of raw material	✓		
3. Transport facilities	✓	✓	
4. Infrastructure availability	✓	✓	
5. Labour and wages	✓	✓	
6. Law and taxation	✓	✓	
7. Suitability of land and climate	✓	✓	
8. Supporting industries and services		✓	✓
9. Community and labour attitudes		✓	✓

Evaluating Location Alternatives



1. Location Cost-Volume-Profit Analysis

The Cost-Volume-Profit (CVP) Analysis can be represented either mathematically or graphically. It involves three steps:

- 1) For each location alternative, determine the fixed and variable costs,
- 2) For all locations, plot the total-cost lines on the same graph,
- 3) Use the lines to determine which alternatives will have the highest and lowest total costs for expected levels of output

$$\text{Total cost} = FC + v(Q)$$

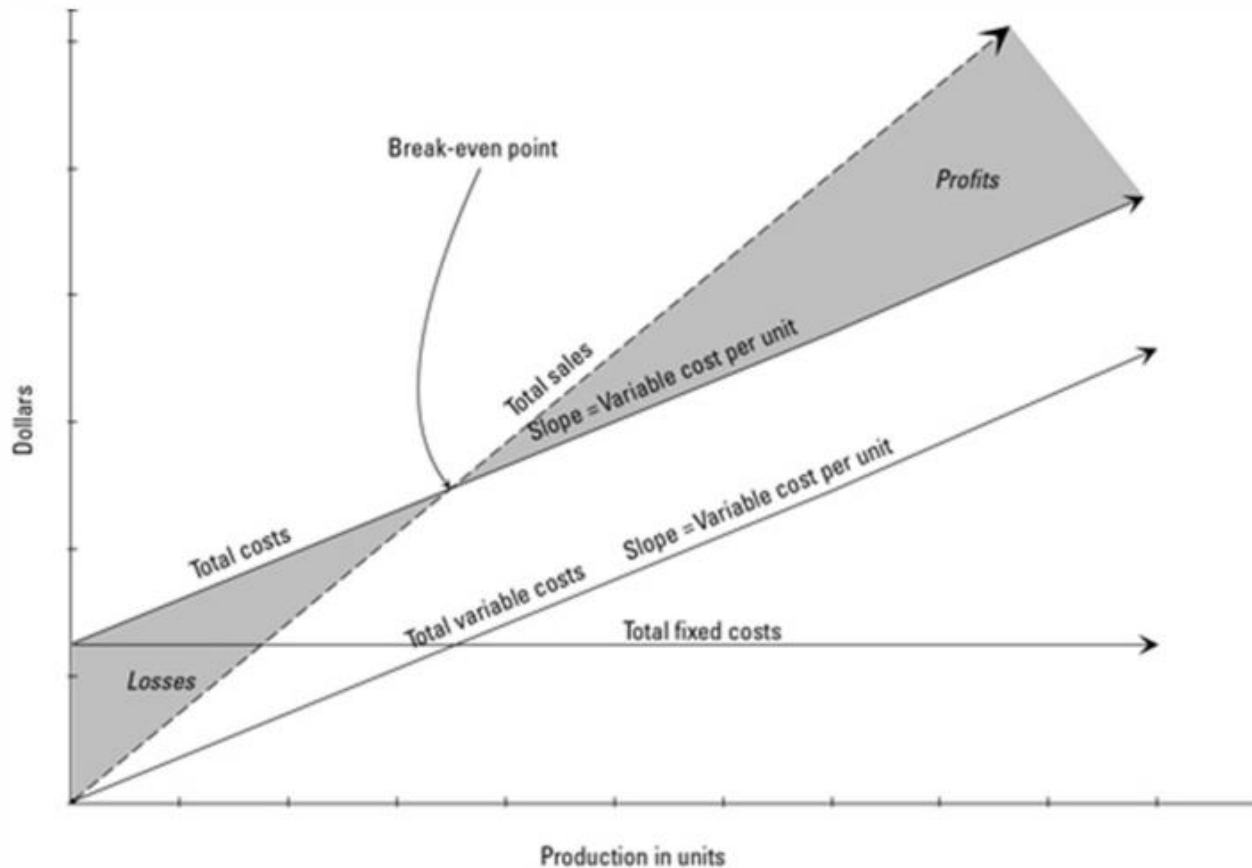
where

FC = Fixed Cost,

v = Variable Cost per Unit,

Q = Number of Units (Also shown below but not in the same format)

Cost Volume Profit Analysis



Fixed and variable costs for four potential plant locations are shown below:

Location	Fixed Cost per Year	Variable Cost per Unit
A	\$250,000	\$11
B	100,000	30
C	150,000	20
D	200,000	35

- Plot the total-cost lines for these locations on a single graph.
- Identify the range of output for which each alternative is superior (i.e., has the lowest total cost).
- If expected output at the selected location is to be 8,000 units per year, which location would provide the lowest total cost?

For a cost analysis, compute the total cost for each location

$$\text{Total cost} = \text{FC} + v \times Q$$

where

FC = Fixed cost

v = Variable cost per unit

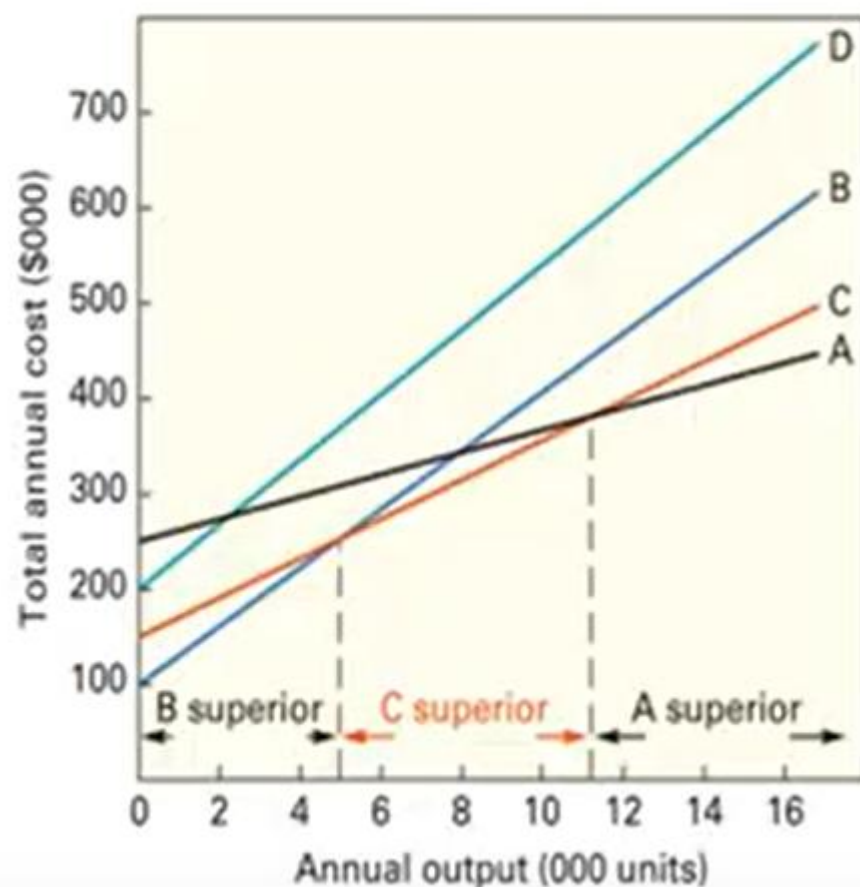
Q = Quantity or volume of output

- a. To plot the total-cost lines, select an output that is approximately equal to the expected output level (e.g., 10,000 units per year). Compute the total cost for each location at that level:

Fixed Cost + Variable Cost = Total Cost

A	\$250,000	+	\$11(10,000)	=	\$360,000
B	100,000	+	30(10,000)	=	400,000
C	150,000	+	20(10,000)	=	350,000
D	200,000	+	35(10,000)	=	550,000

Plot each location's fixed cost (at Output = 0) and the total cost at 10,000 units; and connect the two points with a straight line. (See the accompanying graph.)



- b. The *approximate* ranges for which the various alternatives will yield the lowest costs are shown on the graph. Note that location D is never superior. The *exact* ranges can be determined by finding the output level at which lines B and C and lines C and A cross. To do this, set their total cost equations equal and solve for Q , the break-even output level. Thus, for B and C:

$$\begin{array}{cc} \text{(B)} & \text{(C)} \\ \$100,000 + \$30Q & = \$150,000 + \$20Q \end{array}$$

Solving, you find $Q = 5,000$ units per year.

For C and A:

$$\begin{array}{cc} \text{(C)} & \text{(A)} \\ \$150,000 + \$20Q & = \$250,000 + \$11Q \end{array}$$

Solving, $Q = 11,111$ units per year.

Example 11.4: Break-Even Analysis

- An operations manager is searching for a new facility location to four communities. The annual fixed costs and the variable costs are shown below.

Community	Fixed Costs per Year	Variable Costs per Unit
A	\$150,000	\$62
B	\$300,000	\$38
C	\$500,000	\$24
D	\$600,000	\$30

- Plot the total cost lines for all the sites on a single graph. Identify the approximate ranges for which each location has the lowest cost.
- Solve algebraically for the break-even points over the relevant ranges.
- The company is planning to produce 5,400 units in first year, 10,800 units in second year and 15,000 units in third year. Determine the optimal location for each production plan.

Centre of Gravity Method:

- This technique is used in determining the location of a facility which will either reduce travel time or lower shipping costs
- The Center of Gravity Method involves the use of a visual map and a coordinate system

Plant Layout

- Plant layout refers to the physical arrangement of production facilities.
- According to Moore **“Plant layout is plan of an optimum arrangement of a facilities including personnel, operating equipment , storage space , material handling equipment and other supporting services along with the design of best structure to contain all these facilities.”**

OBJECTIVES OF PLANT LAYOUT

1. Placing equipment in a position resulting in its maximum utilization.
2. Reducing congestion in the flow of materials or movement of workers through successive stages in the process.
3. Providing easy access to equipment for maintenance and repair.
4. Facilitating effective and easy supervision.
5. Minimising production delays.
6. Providing efficient and smooth flow of material within the plant.
7. Providing for fast and safe material handling.
8. Effective utilization of cubic space.
9. Providing comfortable and safe working environment.
10. Providing flexibility to allow minor changes as and when required.

Principle of plant layout

1. Principle of integration - Man, Machine, Material and other supporting facilities for optimum utilization of resources
2. Principle of minimum distance - Minimum travel of man and material
3. Principle of cubic space utilization - That utilize both horizontal and vertical space
4. Principle of flow - Material move in forward direction, there should not be back tracking
5. Principle of flexibility - Altered without much cost and time
6. Principle of safety, security and satisfaction - Give consideration to worker safety and satisfaction and safeguard the machine against fair
7. Principle of minimum handling - Reduce the material handling to minimum

Factor influencing plant layout



- Production type
- Scale of production
- Availability of total area
- Arrangement of material handling system
- Type of building

3.5 PLANT LAYOUT PROCEDURE

Plant layout exercise requires a systematic approach. The procedure requires the following steps:

3.5.1 Information Collection

The following data are to be collected:

- (i) Volume and rate of production.
- (ii) Production specifications and bill of materials.
- (iii) Process sheets indicating tools, equipments and methods.
- (iv) Flow process charts.
- (v) Standard time to complete the operation.

3.5.2 Analysis and Coordination

The information collected is used to decide the following:

- (i) The work force size and type.
- (ii) Number of work stations required.
- (iii) Types of equipment required.
- (iv) Storage and other space requirement.

3.5.3 Choice of Machinery and Equipment

The machinery and equipment needed for the production service is to be decided on the basis of following:

- (i) Number of components to be produced.
- (ii) Capacity of each equipment.
- (iii) Time in which a particular work order is completed.

3.5.4 Selection of Material Handling System

The selection is based on the following factors.

- (i) Material/products to be moved.
- (ii) Container in which it will be moved.
- (iii) Length of movement
- (iv) Speed of movement.

3.5.5 Sketching the Plant on the Plot

Building out line, roads, storage and service areas, etc. are marked on the available plot. It must be kept in mind that the orientation of plant should utilize maximum natural head, light and other weather conditions.

3.5.6 General Flow Pattern

The next step is laying out the details of shop areas as follows:

- (i) Machinery may be laid as per production process requirements, and plant building be constructed around the same.
- (ii) Flow pattern of materials should be such that the distance involved is least between the store and shipping department through the production centers. There should be minimum backtracking and bottleneck.
- (iii) Flow pattern may be analysed using operation process charts or flow process charts.

3.5.7 Individual Work Stations Design

Each work station is laid for achieving optimum performance of operations, material and space utilization and safety and comfort of employees.

3.5.8 Assembly of the Individual Work Station Layout

The total assembly is made in accordance with the general flow pattern and the building facilities.

3.5.9 Storage Space Required

Storage space can be calculated by knowing volume of each stored item, number of items to be kept in stores and the time for each item to be kept in store.

3.5.10 Preparation of Flow Diagrams for Work Stations

After making flow diagrams the work stations are allocated to their respective areas of the plan.

3.5.11 Planning and Locating Service Areas

Service areas such as offices, toilets, wash rooms, rest and lunch room, cafeteria, dispensaries, power generation area, parking areas etc. are planned and located appropriately.

3.5.12 Making Master Layout

Master layout is prepared using templates and models.

3.5.13 Checking Final Layout

The final layout is checked for the safe and economical material handling, adequate production and production control, plant building and its surroundings, service areas, employees safety and comfort, etc.

3.5.14 Approval of Layout

The finally checked layout has to be got officially approved from the concerned authorities.

Plant Layout Problem



- Change in the production design and introduction of new product
- Change in the volume of demand
- Increasing frequency of accident because of existing layout
- Plant and machinery becomes outdated
- Poor working environment affecting worker efficiency and productivity
- Change in location or market
- Minimizing the cost through effective facilities location

Advantages of plant layout



Advantage to the worker

- A good layout will reduce the effort of the workers and minimizes the manual material handling. It reduces the number of accidents and provide better working conditions.

Advantage to the Management

- Effective plant layout reduce the labor costs and enhances the productivity thus ultimately reducing the cost per unit. This helps the management to gain competitiveness in manufacturing.

Advantages of plant layout..



Advantage to the Manufacturing

- Minimizes the movement between work centres and also results in reduced manufacturing cycle.

Advantage to the Production control

- A good layout facilitates production through uniform and uninterrupted flow of materials and helps to carry out production activities within the predetermined time period and with effectiveness.

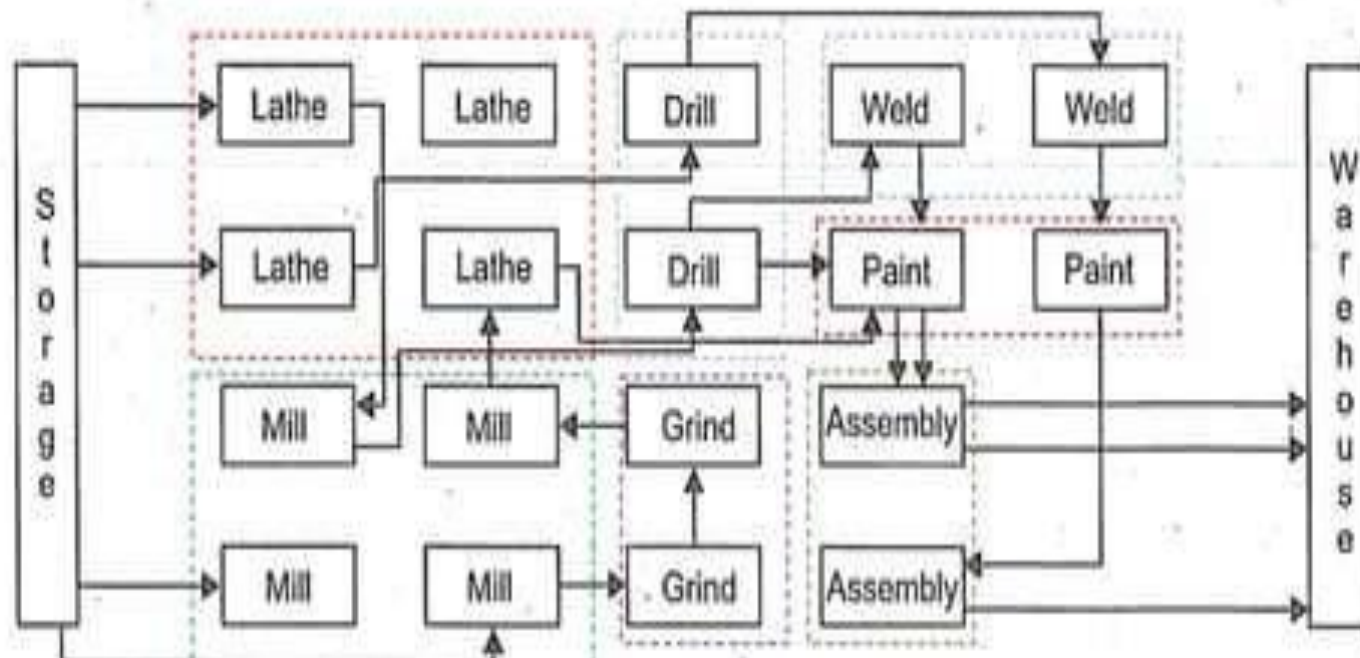
Types of layout

- **Process (Functional) Layout**
- **Product (Line) Layout**
- **Fixed Position Layout**
- **Cellular Layout**



Process Layout / Functional Layout

- In the Process Layout, machines are **grouped according to function to machine centers**. Orders for individual products are routed through the various machine centers to obtain the required processing



Process Layout / Functional Layout



- If a manufacturing facility produces a variety of products in relatively small **batches** will use a process layout
- Process layout typically use **general purpose machines** that can be changed rapidly to new operations for different product designs.
- Machines are arranged according to the type of process being performed
- Workers must be **highly skilled**

Advantages

- Wide **flexibility** exists as regards allotment of work to equipment and workers.
- Better **utilization** of the available equipment.
- Comparatively less number of machines are needed, thus involving reduced capital investment.
- Better product **quality**, because the supervisors and workers attend to **one type** of machines and operations.
- Varieties of jobs coming as different job orders make the work more interesting for the workers.

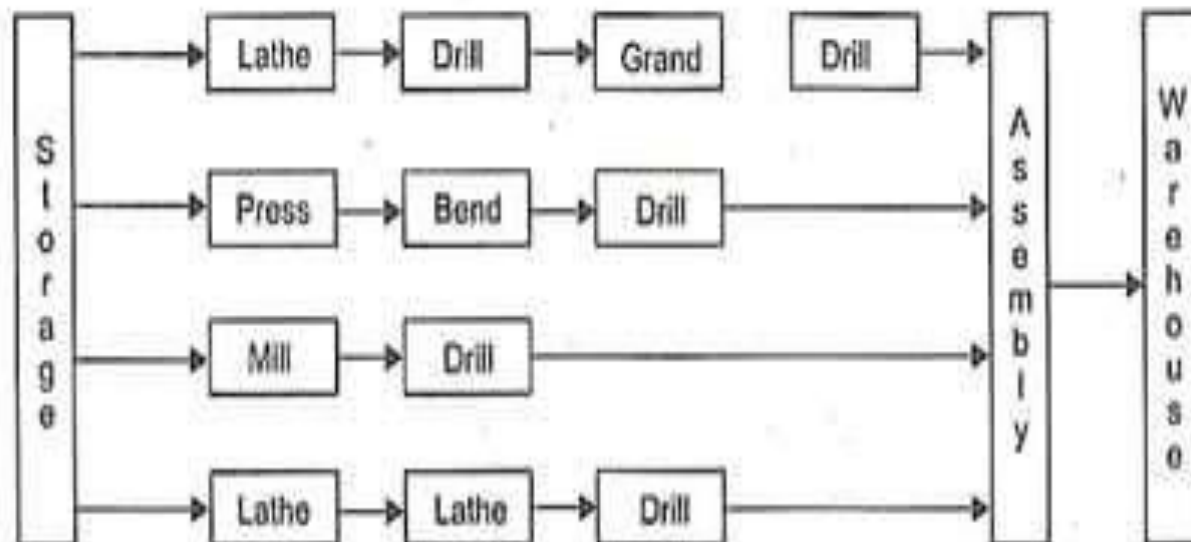
Disadvantages

1. Reducing material handling efficiency due long movement.
2. Material handling can not be mechanized which adds to cost.
3. Process time is long which reduce the inventory turnover and increases the investment in inventories.
4. Production planning and control is difficult.
5. More space is required.
6. Lowered productivity due to number of set-ups.

Product Layout / Line Layout



- The product (or products) follows a fixed path through the production resources. The resources are arranged to minimize the material movement. This type of plant layout is useful when the production process is organized in a continuous or repetitive way



Product Layout / Line Layout



- Product layout is often called **Production line or assembly line**
- Use **specialized machines** to perform specialized operations.
- Changeover takes greater expenses and long down times.
- The amount of **skill**, training and supervision required is **small**.
- Planning and scheduling activities associated with these layouts are complex.

Advantages

- **Less space** requirements for the same volume of production.
- **Automatic material handling**, lesser material handling movements, times and costs.
- Less in-process **inventory**.
- Product completes in lesser time.
- Better co-ordination and simple production planning and control.
- Smooth and **continuous** work flow.
- **Less skilled** workers may serve the purpose.

Disadvantage

- **Change in product** involves **major changes** in layout and thus the layout flexibility is considerably reduced.
- The pace or rate of **working** depends upon the output **rate of the slowest machine**. This involves excessive **idle time** for other machines if the production line is not adequately balanced.
- It is **difficult to increase production** beyond the capacities of the production lines.

Comparison between Product and Process Layout



Characteristic	Product Layout	Process Layout
1. Nature	A sequence of facilities as per processing requirement of products	All similar facilities are grouped together
2. Application	High volume, few products	High volume and high variety
3. Product	Standardised, stable rate of output	Diversified or variety of products using common operations
4. Work flow	Straight line, same sequence of operations for all product	Variable flow for each product type
5. Material Handling	Flow Predictable and systematic can be automated easily	Cannot be automated as flow depends upon the product type
6. Inventory	High turnover of raw materials and WIP inventory	Low turnover of both raw material and WIP
7. Breakdowns	Breakdown in any one machine stops production line	Can tolerate breakdowns
8. Production centre	Simple	Complex
9. Flexibility	Low	High
10. Space utilisation	Efficient	Low
11. Product cost	High fixed cost, Low variable cost	Low fixed cost, High variable cost

Fixed Position Layout



- For tasks on large objects such as the manufacture of an electrical generator, the construction of a building, or the repair of a large air plane, the machines implementing the operation must come to the product, rather than the product moving to the machine
- Example: Missile assembly, large aircraft assembly, ship construction, and bridge construction.

Advantages

1. Material movement is reduced, minimizes damage or cost of moving.
2. Promotes job enlargement by allowing individuals or teams to perform the whole job.
3. Continuity of operations and responsibility results from team. This reduces the problems of re-planning and instructing people each time a new type of activity is to begin.
4. Highly flexible; can accommodate changes in product design, product mix, and product volume.
5. Independence of production centers allowing scheduling to achieve minimum total production time.

Disadvantages

1. Increased movement of personnel and equipment may be expensive.
2. The necessary combination of skills may be difficult to find and high pay levels may be necessary.
3. Equipment duplication may occur.
4. Higher skill requirements for personnel as they are involved in more operations.
5. General supervision required.

Cellular layout or Group layout



- Cellular Layout or Group technology use the technique of identifying and **bringing together related or similar parts in a production process** in order to utilize the inherent economy of flow production methods. Cellular layout is a type of layout in which machines are grouped into what is referred to as a cell.
- In Cellular manufacturing, machines are grouped into cells, and the cells function, like a product layout island within a process layout.

